Risk assessment and mitigation

Group 9 - Inside No.9:

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Description and Justification

Our risk assessment and mitigation process had four stages.

Risk Identification

First we identified risks that the project may have. We derived categories of risks to ensure that our risk register is easily legible and that the span of our identified risks secures our project as much as possible. Categories:

1) Physical Health.
2) Mental Health.
3) Time Constraints.
4) Scheduling Constraints.
5) Software Issues.
6) Hardware Issues.

Risk Analysis

In the second stage we investigated the individual risks further. This was done via the assessment of how likely a risk was to occur and what the impact would be if it did occur. The likelihoods and impacts were rated on a scale from one to ten, where ‘1’ denotes least significance and ‘10’ denotes the most significance. For each risk, a severity rating was calculated by multiplying the likelihood and the impact. The severity rating allows for a shorthand, easy way of understanding a risk for stakeholders.

Risk Planning

Thirdly, we used two categories of risk planning strategies: Avoidance and Mitigation.

Following avoidance strategies, we minimised dependencies on individuals and moved towards team working. We also avoided creating dependencies on any high volatility goals. If any work was dependent on high volatility goals, we ensured that it was handled as a team process to make sure the problem was well covered.

Our mitigation strategies also involved close team working. However, some areas of our code inevitably depend on an individual's knowledge. Our strategy was to have individuals inform the team of their implementation progress during meetings, ensuring everyone stays informed and can research the code further.

More specific strategies are outlined in the Risk Register in the Avoidance / Mitigation column.

Risk Monitoring

Once planning was concluded, we monitored the progress of the outlined risks and re-assessed the likelihood and impact if circumstances changed. We also reassessed any other relevant information for each risk, including our strategies, risk ownership, and monitoring status (either Unencountered, Encountered, or Resolved).

There is an additional data document to aid with colour gradients and severity calculations; **Risk Register Data**

Risk ID	Categorisation	Risk Description	Likelihood	Impact	Severity	Avoidance / Mitigation	Risk Owner	Monitoring Status
1	Physical Health	Eye strain	1	1	1	Ensure all members take regular breaks and observe the “20-20-20” rule.	Individual	Encountered
2	Physical Health	Repetitive strain injury (RSI)	1	2	2	Ensure members take regular breaks and recommend using ergonomic mouse/keyboard	Individual	Encountered
3	Mental Health	Burnout	3	4	12	Ensure equal distribution of work and regularly check in as to whether members are feeling overwhelmed.	Individual	Unencountered
6	Scheduling Constraints	Schedule delays	3	3	9	Keep a thorough timetable and factor in a reasonable amount of slack time to account for potential delays	Oliver Quantrill-Scott	Encountered
8	Scheduling Constraints	Unavailable team member(s)	5	3	15	Ensure there is no task that is only given to a single individual so it can be worked on if that member is unavailable	Sky Burrows	Encountered
5	Hardware Issues	Loss or destruction of hardware	1	5	5	Ensure all hardware is handled carefully and securely and that all documents are stored on the cloud	Individual	Unencountered
4	Software Issues	Software interactions (making versions match)	2	3	6	Ensure all team members are present when implementing software so all can check if there is difficulty with interfacing.	Sky Burrows	Encountered

7	Software Issues	Integration issues	4	3	12	Ensure that members review the structure of relevant pushes so that the project elements can interface with one another more easily.	Erikas	Unencountered
9	Software Issues	Loss of software	1	5	5	Ensure all software is backed up to the cloud and that all previous file versions are kept as to return to them if necessary	Team	Unencountered